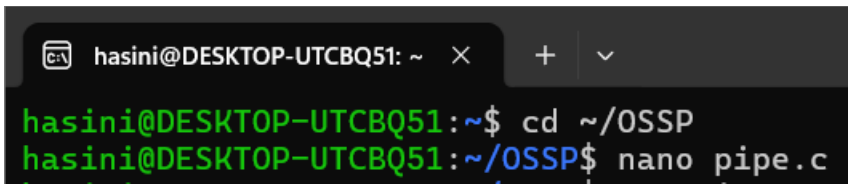
Name: J.chandra hasini
Roll_No:2520030086
Sub: OSSP practical 5b

Develop a program that executes the equivalent of the shell command: ls -l | grep ".c" using fork(), pipe(), dup2(), and exec() system calls.

1.




```c
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    int fd[2];

    pipe(fd);

    if(fork()==0)
    {
        dup2(fd[1], STDOUT_FILENO);
        close(fd[0]);
        close(fd[1]);
        execlp("ls","ls","-l",NULL);
    }

    if(fork()==0)
    {
        dup2(fd[0], STDIN_FILENO);
        close(fd[0]);
        close(fd[1]);
        execlp("grep","grep",".c",NULL);
    }

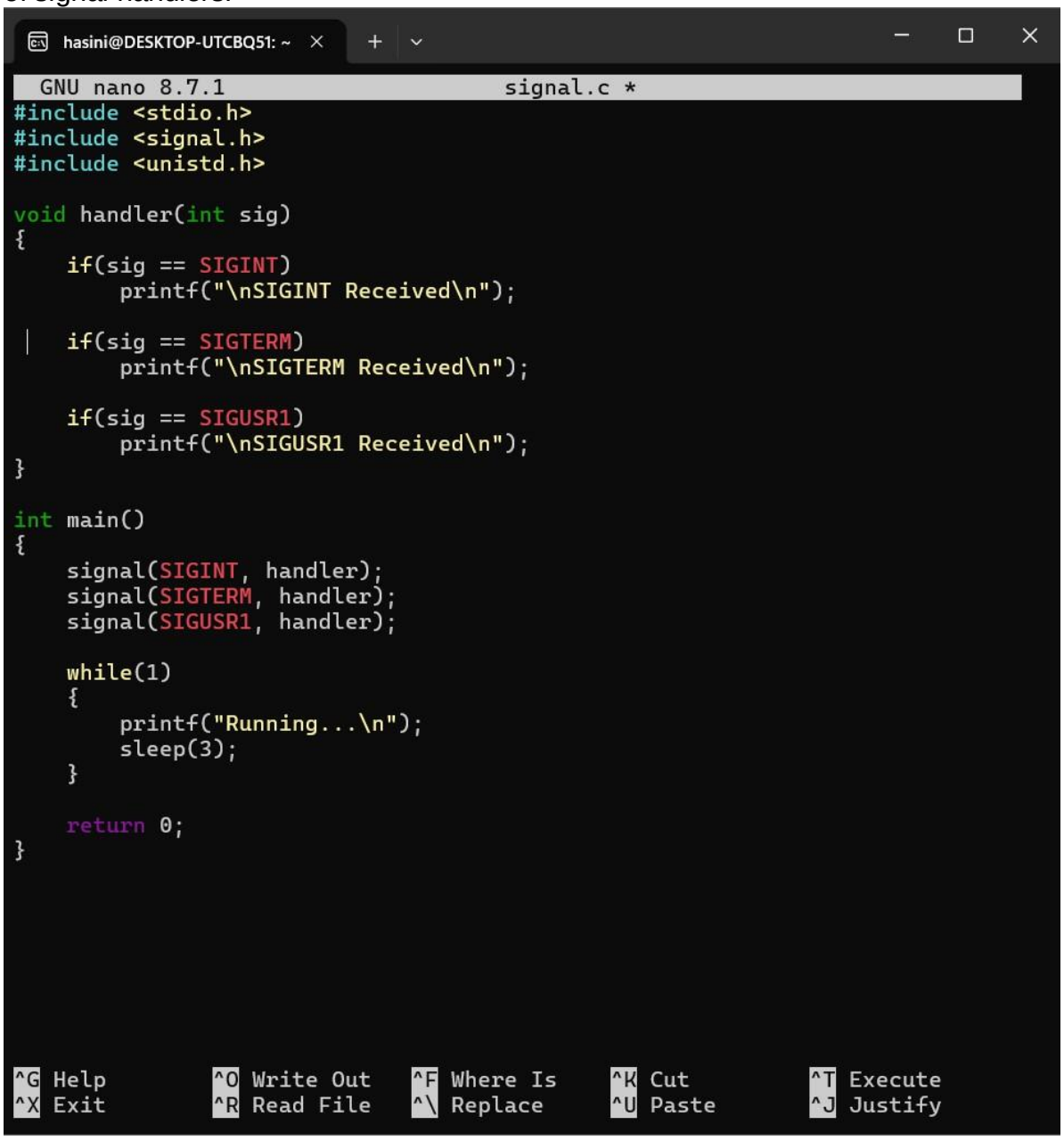
    close(fd[0]);
    close(fd[1]);

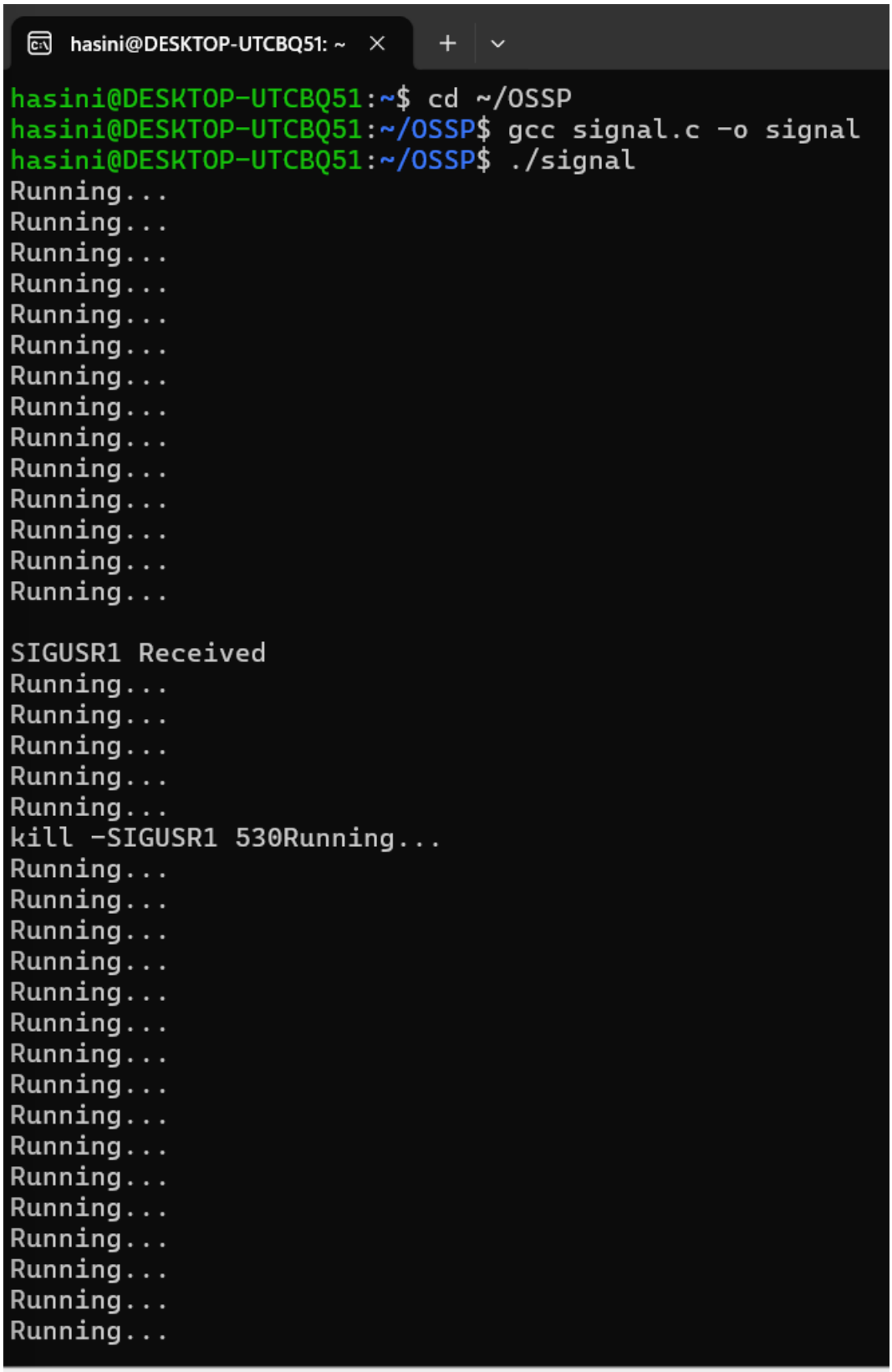
    wait(NULL);
    wait(NULL);

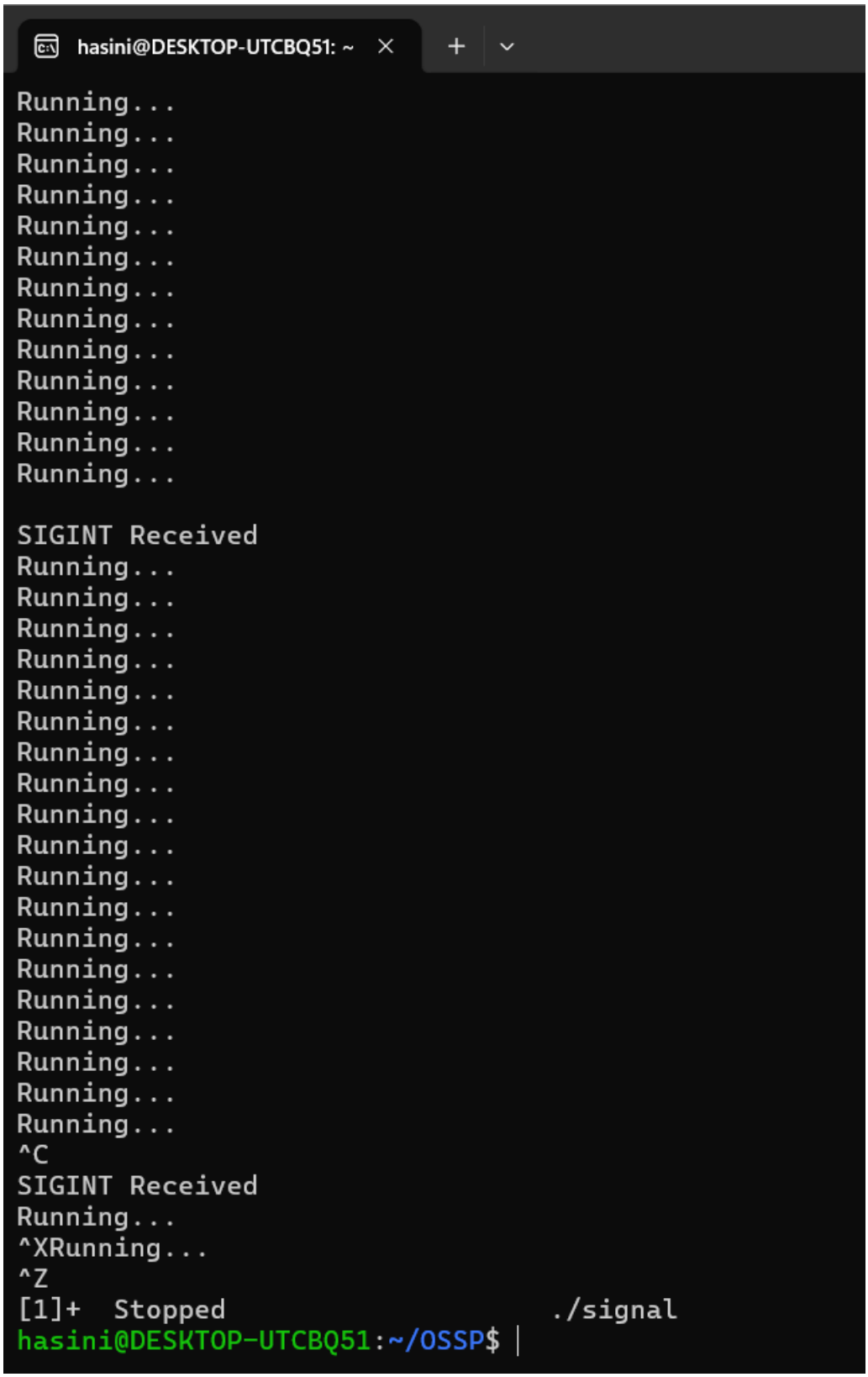
    return 0;
}
```

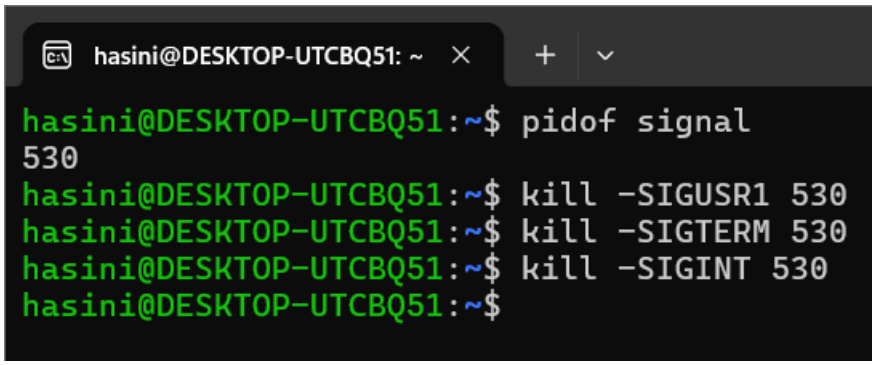


2. Create a POSIX signal handling program that captures SIGINT, SIGTERM, and SIGUSR1. Demonstrate asynchronous event handling and explain the role of signal handlers.




hasini@DESKTOP-UTCBQ51: ~
hasini@DESKTOP-UTCBQ51:~$ cd ~/OSSP
hasini@DESKTOP-UTCBQ51:~/OSSP$ gcc signal.c -o signal
hasini@DESKTOP-UTCBQ51:~/OSSP$ ./signal
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
SIGUSR1 Received
Running...
Running...
Running...
Running...
Running...
kill -SIGUSR1 530Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...


hasini@DESKTOP-UTCBQ51: ~
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
SIGINT Received
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
^C
SIGINT Received
Running...
^XRunning...
^Z
[1]+ Stopped ./signal
hasini@DESKTOP-UTCBQ51:~/OSSP$


hasini@DESKTOP-UTCBQ51: ~
hasini@DESKTOP-UTCBQ51:~$ pidof signal
530
hasini@DESKTOP-UTCBQ51:~$ kill -SIGUSR1 530
hasini@DESKTOP-UTCBQ51:~$ kill -SIGTERM 530
hasini@DESKTOP-UTCBQ51:~$ kill -SIGINT 530
hasini@DESKTOP-UTCBQ51:~$